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VAPT FELLOWSHIP

TASK # 2

TASK 1 – Purpose Of Wireshark

Wireshark is an open-source network protocol analyzer that enables us to analyze, observe captured network traffic either in real time or from saved capture data files. This can prove essential in troubleshooting issues and for determining network behaviour. Also, we can detect malicious activities through the captured data packets.

TASK 2 – How To Approach A Packet Capture

In order to analyze a packet capture we use the following methodology:

- First we load the **.pcap** or **.pcapng** file into Wireshark.
- Then we can apply different filters to filter the traffic like filters for protocols (http, dns, tcp), IP addresses (ip.src = 192.168.23.45) or for specific ports (tcp.port == 80)
- Then we can move to inspect packet details like packet lists, packet details and packet bytes.
- Further analysis can be done to deep dive into endpoints and for generating traffic trends etc.
- We can also right click a packet and follow the TCP Stream to reconstruct full communication sessions, which is also useful for spotting credentials or anomalies.
- Then after following a specific TCP, UDP or HTTP stream we can look for unexpected protocols, ports or traffic spikes.
- After doing all necessary analysis and investigation we can save filtered data, export packet bytes or generate reports for documentation.

TASK 3 – Protocol Deep Dive (DNS)

Domain Name System (DNS) is an application-layer protocol that translates domain names (eg, example.com) into IP addresses. It basically uses UDP on port 53, with TCP as a fallback for large responses.

Message Structure:
Each DNS message consists of five sections:

1. **Header (12 bytes)**
   This contains control information:

   - **ID (16 bits):** Transaction ide to match queries and responses.
   - **QR (1 bit):** Query (0) or Response (1).
   - **Opcode (4 bits):** Type of query (eg, 0 = standard query).
   - **AA (1 bit):** Authoritative Answer (set if the server is authoritative)
   - **TC (1 bit):** Truncated (set if the message exceeds 512 bytes (UDP limit).
   - **RD (1 bit):** Recursion Desired (requested by the client).
   - **RA (1 bit):** Recursion Available (returned by the server).
   - **RCODE (4 bits):** Response code (eg, 0 = no error, 3 = name not found).

2. **Question Section**
   This contains the query:

   - **QName:** Domain name in label format (eg, www.example.com as 3www7example3com0).
   - **QType:** Resource record type (eg, 1 = A, 5 = CNAME, 15 = MX).
   - **QClass:** Usually 1 for Internet (IN).

3. **Answer Section**
   This returns resource records (RRs) that answer query:

   - **Name, Type, Class:** As in the question.
   - **TTL:** Time to live in seconds (how long to cache the record).
   - **RDLength:** Length of RDATA.
   - **RDATA:** Data (eg, IPV4 address for A record, domain name for NS/CNAME).

4. **Authority Section**
   Lists authoritative name servers (NS records) for the domain.

5. **Additional Section**

   Provides “glue” records (eg, A/AAAA records for NS entries) to avoid extra lookups.

Key Behaviours:

- **Name Compression:** Uses pointers to avoid repeating domain names.
- **Recursive vs Iterative Queries:** Clients typically send recursive queries, servers respond recursively or refer iteratively.
- **DNSSEC:** Add security via digital signatures (AC/CD flags, RRSIG records).